SPATIAL AND SOCIAL ASPECTS OF CROWDING PERCEPTION

ANDREW BAUM *is Assistant Professor of Psychology at Trinity College in Hartford, Connecticut. He is interested in the social and psychological aspects of both architectural design and crowding and is associated with theHuman Design Group.*

GLENN E. DAVIS *is a graduate student in the social psychology program at the State University of New York at Stony Brook, and is currently a lecturer in psychology at Trinity College. His interests include architectural design and environmental perception, and he is associated with the Human Design Group.*

Recent studies of human crowding phenomena (Stokols et al., 1973; Valins and Baum, 1973) have stressed that crowding is an experiential variable and is not simply synonymous with high levels of human density. Stokols (1972), in developing a model of human crowding phenomena, has suggested an important conceptual distinction between the physical condition of density with concomitant spatial restriction and the psychological experience of crowding. He has further argued that the perception of spatial restriction or inadequacy may be best viewed as a "necessary antecedent of, but not always a sufficient condition for the arousal of crowding stress." This experience is mediated by two groups of variables: (1) environmental properties, which include both the social and

---

AUTHORS' NOTE: *This research was conducted while both authors were at SUNY at Stony Brook. The assistance of Carlene S. Baum and Paul J. Hopstock during analysis of results and the writing of this paper is gratefully acknowledged.*

Environment and Behavior, Vol. 8 No. 4, December 1976

physical properties of the situation, and (2) personal attributes, such as an individual's past experience (Stokols, 1972). These mediating variables influence the degree to which spatial limitation and unwanted social interaction are perceived as salient and as impinging on the individuals in the setting (Desor, 1972; Stokols, 1972; Valins and Baum, 1973).

These considerations suggest that changes in the physical and social characteristics of a setting will influence the way in which the setting is experienced and, as a result, influence the experience of crowding. Desor (1972) found that by varying architectural features which increased the impact of individuals on one another, substantial differences in perceived room capacities and judgments of crowding could be obtained. By influencing the level of perceived social interaction in model rooms, these architectural variations effectively altered perceived room capacities even though the actual amount of space available was not changed. Similarly, she found that different orientations for the activity in the model rooms also influenced subject perceptions; more figures were placed in the model rooms when the situation in them was defined as social in nature than when the situation was essentially nonsocial. These findings are consistent with Stokols' model of crowding phenomena. The architectural and social manipulations examined in this study may be considered as environmental variables which mediate the experience of crowding by influencing the levels of social stimulation present.

The current research is concerned with the effects of these mediating environmental variables. Specifically, this study examined the effects of room color and visual complexity on perceptions of room capacity and judgments of crowding. Color variation is seen as a physical alteration of the setting which will influence the amount of space perceived available in that setting, while visual complexity is viewed as a physical alteration which will influence the level of social stimulation perceived present in the setting.

COLOR

Most of the psychological research done on color is not contemporary and, for the most part, has focused on subjects' mood associations and preferences for various hues. Others have been interested in evaluating which colors are most appropriate in various climates. Neither of these approaches has adequately dealt with the psychological impact of colors in interiors, but designers acknowledge the importance of color effects in interior spaces (Friedman et al., 1970).

Despite the apparent lack of empirical evidence, designers have been aware that light-colored interiors appear more spacious than darker spaces of comparable size (Pahlmann, 1968). By adding white pigment to a hue, its value is heightened and the intensity and luminosity of the color is increased (Anderson, 1940). It is this increase in brightness which apparently causes a space to appear larger than it would if it were darker and less bright. This suggests that light colors in interior space will cause an individual to perceive it as larger than a comparable space painted with darker colors, and that these differences will influence perceptions of the interior's capacity for people. If an individual perceives a given space as being large, he will be able to tolerate greater numbers of others in that space before feeling crowded, as the greater availability of space perceived will negate experiences of spatial restriction. In contrast, an individual will perceive relatively less space in a dark-colored room. Thus, crowding will be perceived with fewer others present, since the smaller amount of space seen as available is quickly used by others in the setting. Variation of color along these lines should produce differences in perceptions of room capacity and in judgments of crowding.

VISUAL COMPLEXITY

The effect of visual stimulus complexity has been of theoretical interest for years, but has only recently been a subject of discussion among behaviorally oriented architects, designers, and critics of modern public architecture. Studies of sensory deprivation, in conjunction with a reemergence of the cognitive orientation in psychology, have suggested that humans need a certain level of sensory stimulation to function effectively (Hebb, 1955; Brownfield, 1965). The variable of visual stimulus complexity has been characterized in terms of its curiosity or interest value and as an impetus for exploratory behavior (Fiske and Maddi, 1961; Wohlwill, 1968) and has been found to be related to free-looking time (Berlyne, 1960; Gashke et al., 1968). Increased visual complexity of stimulus configurations affords more opportunity for exploration and, as a result, is associated with increased interest and attention.

These findings are consistent with accounts of the character and psychological impact of designed environments. Both Parr (1965) and Mumford (1953) have stressed the possibility of long-range deleterious effects of visual monotony characteristic of modern public architecture. They have indicated that lack of visual complexity leads to a boring and relatively unstimulating cityscape. Similarly, Rapoport and Kantor (1967: 211) suggest that design oversimplification may have unforeseen consequences due to people's perceptual needs:

> A range of meanings and possibilities has been eliminated. This leads to loss of interest—there is nothing to divert or hold one as a result of lowered rates of perceptual inputs.

Visual complexity, in their terms, may be considered as an environmental resource because it activates interest in or curiosity about features of the physical setting and promotes alternative reactions to the same environment.

The importance of these alternative reactions and of individuals' perceptions of them is obvious when considering crowding. When crowding is experienced in the context of high levels of social stimulation and interpersonal encounter, individuals may respond by attempting to reduce the stimulation experienced. By screening or filtering this social input, the individual may be able to restore social experience to desired levels and reduce the arousal of crowding (Milgram, 1970; Desor, 1972; Valins and Baum, 1973). Interior spaces which contain visually complex stimuli provide additional loci of person-environment transaction, increasing the range of acceptable nonsocial behaviors possible. In this manner, visually complex settings provide alternative behaviors for those in the setting, so that they can attend to nonsocial elements and reduce social stimulation experienced. Variation of visual complexity in otherwise comparable spaces will influence the manner in which the space is experienced, and will increase the capacity of the space when social conditions prevail.

It seems likely, then, that both room color and visual complexity are active environmental mediators of the experience of crowding. In order to clarify these assumptions, the current research has attempted to vary these environmental variables in model room settings similar to those employed by Desor (1972). It was predicted that effects of orientation similar to those obtained by Desor would be found; spaces characterized by social activity would have larger preceived capacities than those characterized by nonsocial activity. In addition, it was predicted that spaces painted a dark color would be perceived as smaller and stuffier than those painted a light color, and that subjects would perceive the dark-colored rooms as crowded with fewer figures in them than they would light-colored spaces. Finally, it was predicted that visual complexity would provide alternative behaviors for individuals in the settings, leading to perceptions of increased capacity when compared with settings in which complexity was absent. This hypothesis was qualified by the expectation that increased visual complexity in nonsocial situations would not provide appropriate alternative behaviors, as social stimulation is less of a concern, but instead would provide extra nonsocial stimulation which could increase the likelihood of sensory overload in the settings and reduce effective room capacity. Verification of these predictions will clarify further the nature of environmental mediation of crowding stress and will suggest strategies for dealing with crowding and high density.

METHOD

SUBJECTS

Eighty male and female undergraduates at the State University of New York at Stony Brook participated in this study and received partial course credit in Introductory Psychology as compensation. Prospective subjects were told only that they would be participating in a study of environmental perception.

DESIGN

In order to examine the impact of color, visual complexity, and orientation variables, subjects were asked to consider scaled-down model rooms and small clothespin figures representing people. These rooms were presented randomly, and each subject worked with only one room. In an attempt to reduce variability associated with the nature of the specific task, a factorial design was used in place of repeated-measure designs typically used in studies of this kind.

The three independent variables were crossed in a 2 x 2 x 2 design. Subjects received directions indicating that they were considering either a cocktail party (social orientation) or an airport waiting room (nonsocial orientation), with random assignment to these conditions with the exception that half in each were of each sex. The physical manipulations of color and visual complexity were varied in the model rooms; two of these rooms were painted a light green color, two were painted a dark green, and one light room and one dark room were left devoid of any other features (low visual complexity) while the other two were structured by placing pictures on two walls (high complexity). Again, assignment to these conditions was random with the exception noted above.

Additional data concerning subjects' perceptions of the room were collected, and background data were collected for each subject in order to assess any effects these variables might have had on performance in the experimental situation.

Five college-age women served as experimenters in this study.

PROCEDURE

With few exceptions, the procedures for this study closely followed those used by Desor. Subjects were tested individually and, rather than working with all four rooms, were allowed to see and work with only one. On entering the laboratory, subjects were told that they were participating in a study of environmental perception and were seated in front of a model room and a box of miniature figures. After the basic procedure of considering the model was explained, subjects were given written instructions and told to proceed. The instructions explained the task and supplied the specific orientation for the room activity. The social orientation was conveyed by asking subjects to "imagine that this room is a lounge where people are standing and talking at a cocktail party, and that you are there." Subjects in the nonsocial conditions were asked to consider the room as "a lounge area in an airport where people are standing and waiting between flights, and that you are there." Attention was called to a single colored clothespin figure, representing the subject, and participants were asked to place "themselves" in the room. Following this, subjects were asked to place as many clothespin people in the room "as you can until you begin to feel crowded."

When figure placement was completed, the model room was covered so that it could no longer be seen, and subjects were asked to rate the room along several dimensions. Finally, background information on each subject was obtained by asking each to complete a brief questionnaire, after which the purposes of the study were explained more fully and the session was terminated. At this point, the experimenter transposed the location of each figure placed in the model room onto an in-scale drawing of the room. These data sheets served as the permanent record of figure placement.

THE MODEL ROOM

Four scale model rooms were constructed of plywood, representing rooms 20¼ feet square with walls 8¼ feet high (scale: 18 inches of real distance = 1 inch of model room). The interior walls of the two rooms in the dark-color conditions were painted with Sears paint No. 0872, while those used in the light-color conditions were painted with Sears No. 0866—which was basically the same color lightened considerably y the addition of white and yellow pigments. Two rooms, used in the low visual complexity settings, were left devoid of any other characteristics, while the other two were given increased visual complexity by placing pictures along two walls. Each of these rooms had the same pictures placed in the same positions. One picture, a 2¾ inch x 2½ inch graphic lion, was placed on the back wall 2½ inches from its intersection with the right wall. The other two pictures were placed along the right wall: a painting of a bridge (2¾ inch x 2½ inch) was placed 2½ inches from the intersection of the right and front walls, and a picture of a city street (1½ inch x 2¼ inch) was placed 2½ inches from the back. Rooms were provided with floors made of cork.

A large supply of clothespin figures (3¾ inches tall) representing people 5 feet, 8 inches in height were provided. Pushpins were glued to the base of each figure so that they could stand upright on the cork floors of the rooms. Subjects were seated so that the front wall was always nearest to them and so that the box of miniature figures was to their right.

DEPENDENT VARIABLES

Records of figure placement in the model rooms provided two kinds of information. First, the number of figures placed in each room was interpreted as an indication of subjects' criterion of crowding. As in previous studies, "the addition of one more figure into a subject's display would push the situation over his cut-off between uncrowded and crowded or tolerably crowded and overcrowded" (Desor, 1972). Differences in the number of figures placed in the rooms were interpreted as indications of the effects of manipulated social and physical variables on subject perceptions of crowding and room capacity. In addition, records of figure placement contained information regarding where the figure representing the subject was placed, allowing post hoc analysis along this dimension. The importance of this measure was based on the assumption that where a subject elects to place himself in the model room indicates whether he is using the alternative interaction loci provided by increased complexity. If, for example, subjects place themselves near the pictures when considering social conditions, it may indicate that they are avoiding social contact by attending to nonsocial elements in the situation. If, on the other hand, subjects show no consistent preference for self-placement, it would appear that the alternate loci provided by complexity are not being used in the situation.

Rating of the room following figure placement provided information on the subjects' perceptions of the rooms' prettiness, brightness, warmth, stuffiness, size, comfort, and the degree to which subjects liked the rooms. Subjects also indicated how crowded they felt the room was. All of this information was assessed on seven-point scales (1-7), with higher numbers indicating more of the dimension being considered. This data not only served as manipulation checks of the effect of color on perceived size, but also yielded information used to clarify figure placement data.

Finally, a background questionnaire was administered to each subject at the end of the experimental session. Subjects were asked to indicate their age, sex, Scholastic Aptitude Test (SAT) scores, ordinal position, family size, and current home address. Indices of residential history were obtained: subjects were asked if their childhood environment was most typically characterized by rural, small town, moderate town or city, suburban, or urban settings.

RESULTS

Three-way analyses of variance were performed on most of the data collected, considering the individual subject as the unit of analysis. Since the results were not influenced by the experimenter variable when introduced, statistical analyses are presented collapsed across this variable.

FIGURE PLACEMENT

As was predicted, color differences accounted for a significant amount of the variance in the number of figures placed in each room ($F = 25.116$, $p < .001$): more figures were placed in the light green rooms than in the dark green rooms (see Table 1). However, the visual complexity variable did not account for a significant share of the variance ($F = .836$, $p > .3$), and the effect of orientation reported by Desor was not obtained ($F = .482$, $p > .4$). When the high complexity conditions were considered alone, the interaction of orientation and room color approached significance ($F = 2.72$, $p \simeq .1$); while the number of figures placed in the light green room and the dark green room considered under social orientation was comparable, the number placed in the dark green room considered under nonsocial conditions was considerably lower.

Examination of the position in the room selected for the figure representing the subject reveals additional information. While subjects considering low visual complexity rooms did not exhibit significant tendencies regarding self-placement in the

TABLE 1
Figure Placement in Model Rooms, Orientation by Color, by Visual Complexity

Orientation	Low Visual Complexity		High Visual Complexity	
	Light Green	Dark Green	Light Green	Dark Green
Social	27.00	17.60	29.80	23.60
Nonsocial	28.70	17.30	30.50	15.10

rooms (see Table 2), those considering high complexity rooms tended to place themselves in the right half near the pictures ($X^2 = 6.084$, df = 1, $p < .05$). Closer examination of these data also reveals that these differences were generated primarily in the light green, high complexity room and in the dark green, high complexity room considered under social orientation. Self-placement in the right half of the dark, high complexity room considered under nonsocial conditions was considerably lower than in the other rooms characterized by high visual complexity.

ROOM RATINGS

Analysis of the ratings of the rooms following figure placement demonstrates a strong main effect of color along most dimensions (see Table 3), and interactive effects similar to those observed for figure placement and self-placement. The differences in color did influence the perceived size of the rooms, as subjects considering dark green rooms rated them significantly smaller than those rating light green rooms ($F = 26.259$, $p < .001$). However, an interaction among the three major independent variables indicated that the dark green room characterized by high visual complexity was perceived as smaller than other high complexity rooms when considered under a nonsocial orientation ($F = 15.003$, $p < .001$). Ratings of stuffiness ($F = 15.346$, $p < .001$) and the degree to which crowding was perceived in the rooms ($F = 10.544$, $p < .01$)

TABLE 2
Percentage of Subjects Placing Figure Representing Themselves in the Right Half of the Model Rooms (right-half placement indicates placement near pictures)

	Low Visual Complexity		High Visual Complexity	
Orientation	Light Green	Dark Green	Light Green	Dark Green
Social	40	40	80	70
Nonsocial	40	50	80	50

were also influenced by room color and showed the same three-way interactions as did ratings of room size ($F = 5.389$, $p < .05$ and $F = 5.816$, $p < .05$).

There were no observed differences along any manipulated dimensions for ratings of the rooms' prettiness, brightness, or warmth. However, a significant sex effect was found for ratings of how much subjects liked the rooms: female subjects indicated that they liked the rooms more than males did ($F = 7.031$, $p < .05$). The absence of any effects of room color, visual complexity, or social orientation on the ratings of prettiness and attraction indicated that these manipulations did not alter aesthetic ratings of the rooms or influence preference for them.

BACKGROUND CHARACTERISTICS

Personal characteristics, including age, family size, ordinal position, and residential history, were not related to any of the dependent variables; correlations for these relationships did not exceed .25. It should be noted, however, that the sample used in this study was relatively homogeneous. More than 80% of the subjects reported that they had been raised in Metropolitan New York, and the ages and family histories of the subjects were similar. It is likely that this homogeneity effectively limited the possibility of observing differences along these dimensions.

TABLE 3
Ratings of the Model Rooms Following Figure Placement by Room Color

	Rating Dimensions					
Room Color	Pretty	Bright	Cool	Stuffy	Small	Crowded
Light Green	2.90	2.46	2.85	2.17	2.02	2.25
Dark Green	2.90	2.42	3.47	3.52	3.50	3.32

DISCUSSION

This research was primarily concerned with environmental mediation of perceptions of crowding. It was predicted that light-colored interior spaces would appear larger to subjects and would therefore accommodate more miniature figures before becoming crowded than would dark-color rooms. Similarly, it was predicted that visual complexity would influence interior capacity; added interest and the alternative responses provided by environmental complexity would divert attention from other people in a space, reducing the impact and saliency of their presence. This prediction was further qualifie by the expectation that added visual complexity would be most effective in depressing crowding perceptions in social situations where interacting others are of primary concern. It was specifically noted that dark rooms characterized by high visual complexity and essentially nonsocial activities would be perceived as smaller and more restrictive than various other spaces because of additional sensory stimulation which could impinge on an individual's processing operation and cause overload. Finally, it was predicted that Desor's (1972) finding that social orientation influenced room-capacity perception would be replicated.

The data generally confirmed these expectations, although no main effects for complexity or orientation were found for figure placement or for room ratings. However, consistent interactions among the three independent variables verify the principles assumed active in these settings. While failure to replicate Desor's findings regarding social orientation may be due to the fact that only two of the four activities she used were considered in this study, it also appears that this variable may be maximally effective in determining capacity and crowding judgments when repeated measures designs are used. The use of a wider range of orientations may have increased the influence of social and nonsocial activity in this study by increasing the normative differences between the "most" social and the "least" social setting. On the other hand, repeated measures designs typically used in studies of model rooms may increase the importance of the nature of activity variable by providing each subject with the opportunity to work with both social and nonsocial settings. As a result, subjects examined in this manner can evaluate and compare the norms of interaction fostered by each orientation, and figure placement may become a reactive index of room capacity and perceptions of crowding. The factorial design employed in this study did not allow subjects the opportunity for comparison of these norms, reducing variability associated with the task. Although it is clear that differences in the nature of a setting and among the goals projected into a setting are associated with changing appropriateness of others and perceptions of crowding, the use of model rooms to study varying criteria for crowding associated with these variables may not be sufficient.

Of greater interest was the findings that variation of the environmental properties of a setting can influence individuals' perceptions of model room size and crowding. Dark-colored rooms were perceived as smaller, stuffier, and more crowded than light-colored rooms of the same size, and fewer figures were placed in these dark rooms prior to a judgment of crowding. These differences are not attributable to changes in the attractiveness of the setting, but to the manner in which color influences perceptions of the extent of interior space. Light colors apparently cause interiors to appear larger, increasing the amount of space perceived as available for use by those in the setting. As a result, it appears that more people can comfortably use the setting, and the threshold of crowding will be increased. Dark colors, on the other hand, seem to result in perceptions of decreased spatial availability which lower the threshold for crowding and cause individuals to perceive the setting as being crowded with fewer others present.

Visual complexity is also related to the manner in which interior model-room spaces are perceived, but the manner in which high complexity influences crowding is related to the color of the space being considered as well as the nature of the ongoing activities in the setting. Subjects tended to place themselves near the pictures in the model rooms when they were provided, reaffirming the acknowledged role of visual complexity as an impetus for exploration. In addition, complexity influenced perceptions of the dark rooms: high complexity in these rooms increased capacity and crowding thresholds when the activities in the room were social, but decreased the rooms' capacities and crowding thresholds for nonsocial activity. By providing visually interesting aspects of the situation to which individuals might attend, visual complexity may have provided a plausible means by which individuals could reduce the amount of unwanted and excessive social stimulation they were experiencing. In social settings, interactions with others are more salient than spatial availability in the determination of crowding, and by providing additional loci for interaction, visual complexity may allow individuals to attend to nonsocial information and input rather than social experiences. This may enable individuals to screen themselves from the stimulation levels created by large numbers of people; they may divert their attention from this potential source of stress.

However, when the nature of the activity in the space was not social, perceptions of spatial extent and room capacity were reduced. The combination of dark color, which limits the extent of space perceived to be available, and additional nonsocial stimulation provided by the visual complexity seem to have influenced subjects' perceptions in ways which reduced their thresholds for crowding and led them to feel that the space was crowded with relatively few others present. The primary concern in nonsocial settings is not social interaction but it probably related to spatial limitation and interference, and the visual complexity in this situation no longer served to reduce excessive social stimulation. Instead, it may have added to the levels of nonsocial stimulation already present, approaching subjects' thresholds for sensory overload. In other words, the added stimulation afforded by the visual complexity could have resulted in too much general stimulation in the setting, reducing perceptions of room size and number of others who might comfortably interact in the setting.

While this explanation seems reasonable, it is also possible that subjects were responding to the normative properties of the model-room settings engendered by the particular manipulation of complexity employed. Rather than responding to complexity, a room with pictures may have seemed more suitable for a cocktail party than a room without pictures. Subjects, therefore, may have viewed the high-complexity rooms as more accommodating to larger groups because of their suitability for social activity. In contrast, the nonsocial activity in this study was described as occurring in an airport lounge, where pictures are not often present. Further research should concern itself with this issue and attempt to avoid potential confounds of orientation with more normative expectations.

These findings have unavoidable implications for both the study of crowding and the design of interior spaces. It is clear that intervening variables such as color or environmental complexity are active in the crowding process. Density is defined in terms of those conditions which generally accompany it, and the perception of these conditions as impinging on an individual will typically arouse the experience of crowding. However, the degree to which these conditions are realized and made salient by the characteristics of the setting reflects the likelihood that individuals will perceive the setting as crowded. If, as Stokols (1972) indicates, crowding is experienced when there are perceived disparities between an individual's needs for space and that which is available to him, any combination of situational variables that reduces the amount of space perceived available or that increases perceived spatial requirements will influence this experience. In the present study, environmental properties were varied so that individuals' perceptions of the extent of space were reduced, and this reduction of spatial availability seemed to cause such disparities with fewer others present. These findings provide evidence of the intervention of situational variables in the experience of crowding.

It should be noted, however, that the generalizability of these findings are limited by the fact that subjects in model-room studies do not have any real experience in the setting. By asking subjects to project themselves into a situation, visual properties of the setting assume artificial importance as compared with other forms of stimulation. Olfactory and tactile sensations present in actual experience, for example, cannot be considered active in the model-room experience, and the absence of such cues may exaggerate the importance of the visual cues of color and complexity. Since the experience of crowding is an interpersonal process, the elimination of these sensations and more directly experienced interaction demands may have hindered a more directly experienced interaction demands may have hindered a more realistic examination of environmental mediation of crowding. As a result, these findings should be considered with caution until they are replicated and extended in actual settings where interpersonal interaction is more complete and not artifically limited.

In terms of design, these findings suggest ways of maximizing individual perceptions of spatial extent and of effectively increasing comfortable and uncrowded room densities. By varying color, designers can increase the number of individuals who may comfortably interact in an interior space. However, these findings also indicate the importance of considering as many elements of a setting as possible when design decisions are made; perceptions of crowding and room capacity are necessarily influenced by the juxtaposition of environmental elements which define a setting as well as by psychological characteristics which describe those using the setting. Once again, it should be noted that these findings must be replicated in real-life settings before any definitive conclusions can be made. While this study provides potentially useful information regarding the mediation of crowding by design factors, naturalistic examination of this process seems necessary in order to gain a clearer understanding of these issues.

REFERENCES

ANDERSON, A. M. (1940) Syllabus of Design and Color. Champaign, Ill.: Stipes.

BERLYNE, D. E. (1960) "Conflict and information-theory variables as determinants of human perceptual curiosity." J. of Exper. Psychology 53: 399-404.

BROWNFIELD, C. A. (1965) The Brain Benders: A Study of the Effects of Isolation. New York: Exposition Press.

DESOR, J. A. (1972) "Toward a psychological theory of crowding." J. of Personality and Social Psychology 21: 79-83.

FISKE, D. W. and S. R. MADDI Functions of Varied Experience. Homewood, Ill.: Dorsey.

FRIEDMAN, A., J. F. PILE, and F. WILSON (1970) Interior Design: An Introduction to Architectural Interiors. New York: American Elsevier.

GASHKE, J. A., B. L. KINTZ, and R. W. THOMPSON (1968) "Stimulus complexity, free looking time and inspective exploration." Perception and Psychophysics 4: 319-320.

HEBB, D. O. (1955) "Drives and the C.N.S. (conceptual nervous system)." Psych. Rev. 62: 243-254.

MILGRAM, S. (1970) "The experience of living in cities." Science 167: 1461-1468.

MUMFORD, L. (1953) The Highway and the City. New York: Mentor.

PAHLMANN, W. (1968) The Pahlmann Book of Interior Design. New York: Viking.

PARR, A. E. (1965) "City and psyche." Yale Rev. 55: 71-85.

RALL, J., B. PINNER, and J. SCHOPLER (1973) "Physical, social and personal determinants of perception of crowding." Environment and Behavior 5: 421-439.

RAPOPORT, A. and R. E. KANTOR (1967) "Complexity and ambiguity in environmental design." J. of the Amer. Inst. of Planners 33: 210-221.

STOKOLS, D. (1972) "A social-psychological model of human crowding phenomena." J. of the Amer. Inst. of Planners 38: 72-83.

---, M. RALL, B. PINNER, and J. SCHOPLER (1973) "Physical, social, and personal determinants of the experience of crowding." Environment and Behavior 5: 87-115.

VALINS, S. and A. BAUM (1973) "Residential group size, social interaction and crowding." Environment and Behavior 5: 421-439.

WOHLWILL, J. F. (1968) "Amount of stimulus exploration and preference as differential functions of stimulus complexity." Perception and Psychophysics 4: 307-312.